

Julian Hart

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EDUCATION

B.S. in Computer Science, Concentration in Algorithms and Theory

2027 | Storrs

University of Connecticut

- Relevant Coursework: Data Structures, Algorithms and Complexity, Python Development, Systems Programming, Computer Architecture, Cybersecurity, Theory of Computation, Probability Models, AI, OOP, C++

Associate in Computer Science

2024 | Manchester

CT State Community College

- Graduated on the Dean's List.
- Relevant Coursework: Java, Discrete Math, SQL, MongoDB, HTML, CSS, Tailwind, JavaScript

PROFESSIONAL EXPERIENCE

Software Engineering Intern

Jun 2026 – Present

The Walt Disney Company

- Managed APIs using Java and Spring Boot to power live and on demand media distribution for Disney+, ESPN+, Hulu and more. Ensuring reliable content delivery for over 10 million active users.
- Engineered automated data pipelines leveraging AWS (S3, ECS, and Java based Lambdas) to securely streamline the media dataflow between external content providers and end consumers.
- Worked on a centralized observability dashboard using TypeScript to track real time service uptime, providing immediate visibility into the health and performance of the backend infrastructure.

Software Engineering Intern

Jun 2025 – Dec 2025

IPC Systems

- Refactored legacy turret remoting system to use the WAMPD protocol, replacing direct TCP communication and migrating the API layer from C++ to TypeScript within a Tauri architecture.
- Replaced screenshot-based monitoring with VNC integration in Vue + Tauri, enabling interactive remote access and cutting latency by 60%.
- Implemented remote diagnostic command execution via WAMPD SSH, allowing real-time feedback and control through a TypeScript-based frontend.

Web Development Intern

Sep 2024 – May 2026

University of Connecticut

- Designed and maintained backend architecture for university websites, ensuring uptime and handling more than 1000 daily visitors.
- Engineered and optimized RESTful API endpoints, increasing site speed and reliability.
- Automated data migration processes from Airtable using Python, reducing manual effort by 5 hours weekly.

Co-Founder / Front End Software Engineer

Jun 2024 – May 2026

CheckUP LLC

- Developed core user-facing features for a mental health SaaS platform, enabling seamless user registration and activity tracking.
- Partnered with the back-end team to integrate user data and authentication using Clerk, improving onboarding efficiency by 80%.
- Designed and built responsive user interfaces with React and Tailwind CSS, ensuring WCAG 2.1 accessibility compliance and smooth cross-device performance.
- Spearheaded the platform's UX/UI design process, ensuring a smooth and enjoyable user journey.

PROJECTS

Pneumonia Classifier

- Built and trained a convolutional neural network (CNN) in PyTorch to detect pneumonia from chest X-ray images with high accuracy.
- Deployed the trained model as a Flask API and hosted it on Render with real-time image classification support.

Encryption Library

- Built a modular encryption library in C, implementing secure AES and RSA algorithms for symmetric and asymmetric encryption.
- Leveraged low level memory management using pointers and custom data structures to handle key generation, padding, and block cipher operations efficiently.

Video Meeting App

- Developed a scalable meeting application using Next.js and TypeScript, supporting up to 50 concurrent users with seamless cross-platform functionality.

SKILLS

Languages

Java, JavaScript, TypeScript, Python, C, C++, SQL, Assembly (RISC-V)

Databases & Cloud

MongoDB, PostgreSQL, AWS (EC2, S3, Lambda), RESTful APIs, GraphQL

Core Competencies

Algorithms & Data Structures, Machine Learning & AI (Deep Learning, Computer Vision, Model Deployment), Cross-Platform Development, System Design, Accessibility (WCAG 2.1), Agile/Scrum

Frameworks & Libraries

React, Next.js, Node.js, Express.js, Redux, Tailwind CSS, Bootstrap, Flask, PyTorch, scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib

Developer Tools

Git, GitHub, Linux (Ubuntu), Docker, Postman, VS Code